Limits - General ideas

1 Domain! 2 Substitution 3 Limit laws 4 Draw simple graphs

Evaluate limit or determine form

5 Form $\sqrt[n]{0}$ (n even)

Reason out left and right limit separately

- Domain NB

6 Form $\frac{0}{0}$

Make a plan (Can not be reasoned!)

- Factorise and simplify
- Multiply by 1 (e.g. $\times \frac{x}{x}$ or $\frac{conjugate}{Conjugate}$)
- Special limits
- Trig identities
- L'Hospital's rule

7 Form $\frac{a}{0}$ ($a \neq 0$) (VA if $x \to constant$)

Reason out left and right limit separately

- Possible answers: $\infty, -\infty, \nexists$
- Draw simple graphs if necessary

9 Form $\frac{\infty}{a}$ (a finite) (VA if $x \to constant$)

Reason out left and right limit separately

- Possible answers: $\infty, -\infty, \nexists$
- Draw simple graphs if necessary

10 Form $\frac{a}{\infty}$ (a finite)

- $\frac{a}{\pm\infty} = 0$

11 Form $\frac{\infty}{\infty}$

Make a plan (Can not be reasoned!)

- Factorise and simplify
- Multiply by 1
- Write in form $\frac{0}{0}$
- L'Hospital's rule

12 Form $0 \times \infty$

- Write in form $\frac{0}{0}$ or $\frac{\infty}{\infty}$

8 Form $\frac{f}{\infty}$ or $0 \times f$ where $\lim f$ does not exist but f is bounded • Squeeze theorem